

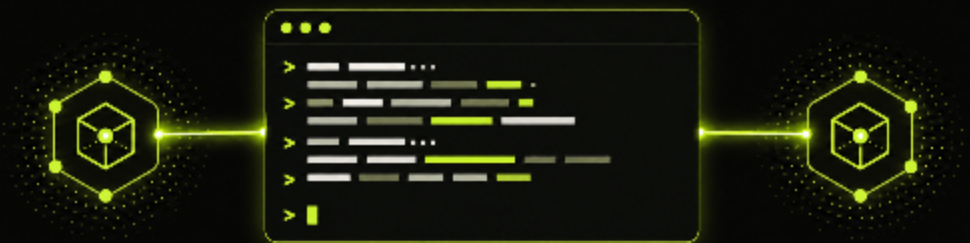
# p2p-netcat

Netcat addressed by PeerId

One cryptographic address for terminal, TCP, UDP, and proxy access across NAT.

## p2p-netcat web

A terminal between two peers.  
Right in your browser.



### IDENTITY

**PeerId instead of IP**

### ROUTE

**First authenticated route**

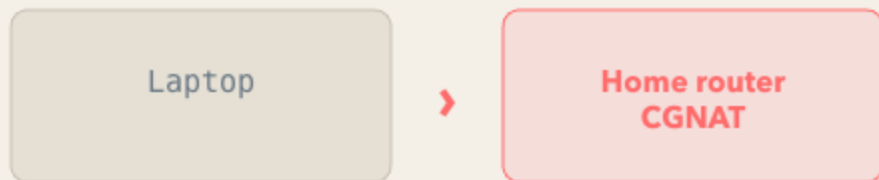
### SESSION

**TCP · UDP · PTY · SOCKS**

# Your home server has no public address – but you still need SSH.

WITHOUT P2P-NETCAT

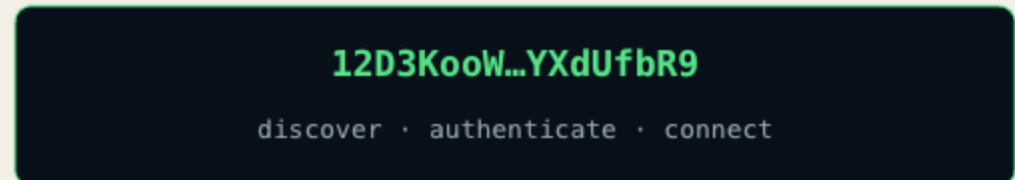
## The network edge is not yours



Port forwarding is unavailable. Dynamic DNS does not help. A hosted VPN adds another control plane.

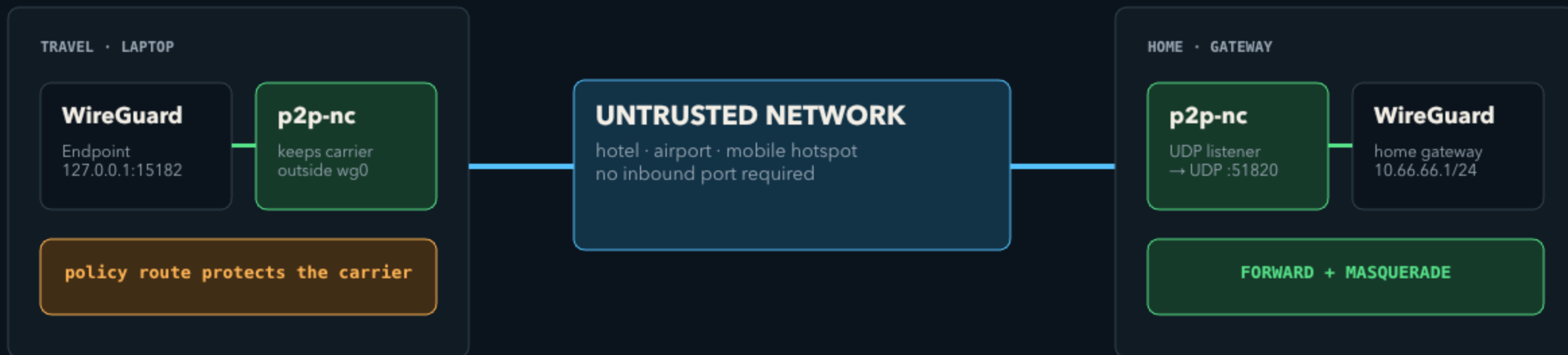
WITH P2P-NETCAT

## The server keeps one identity



Run a listener at home. From anywhere, dial its PeerId and forward local port 2222 to SSH.

# A full Internet tunnel should still work when the gateway is behind NAT.



## ONE ROUTE

bring up carrier



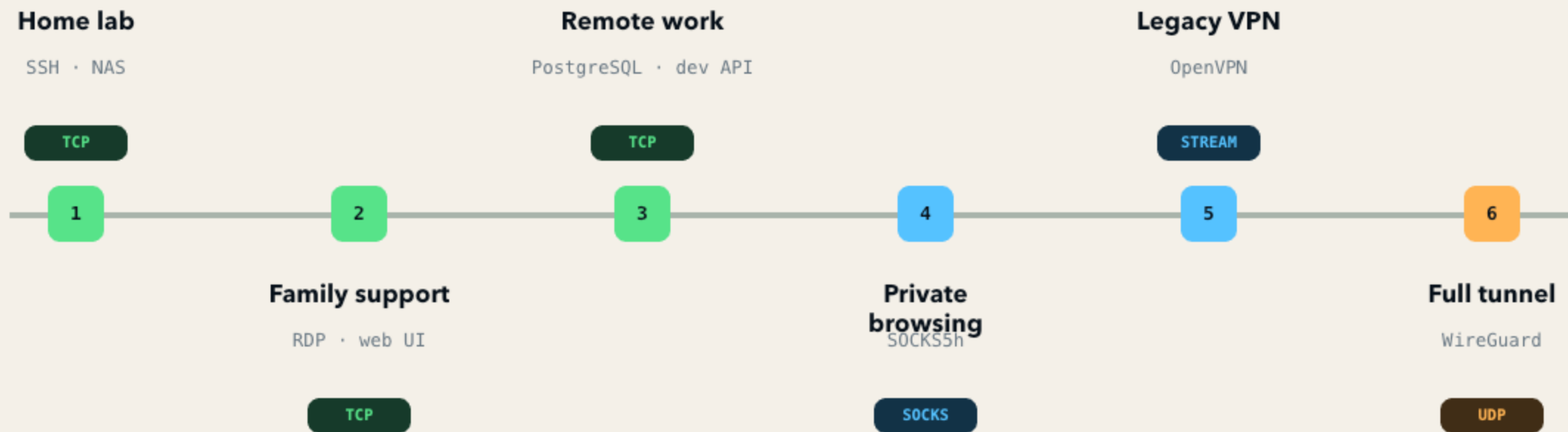
wg-quick up wg0



public IP = home

The P2P carrier reaches the home peer first; WireGuard then routes all Internet traffic through it.

# The same need keeps repeating: reach a private service without owning its network edge.

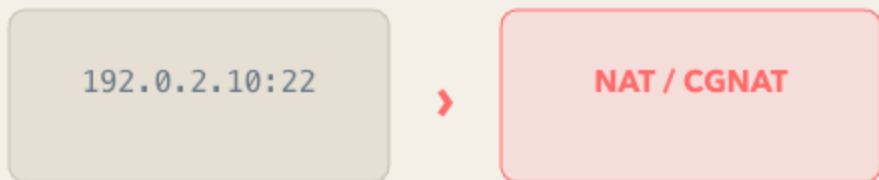


One stable PeerId replaces per-network IP addresses, port mappings, and temporary VPN control planes.

# PeerId solves identity. The route still has to be found.

## CLASSIC FORWARDING

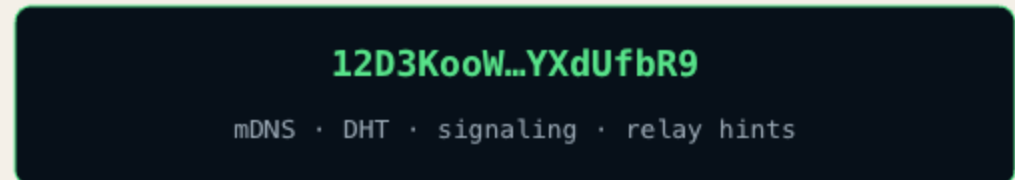
### The IP changes – access breaks



DNS, port forwarding, a VPN concentrator, or manually operated infrastructure.

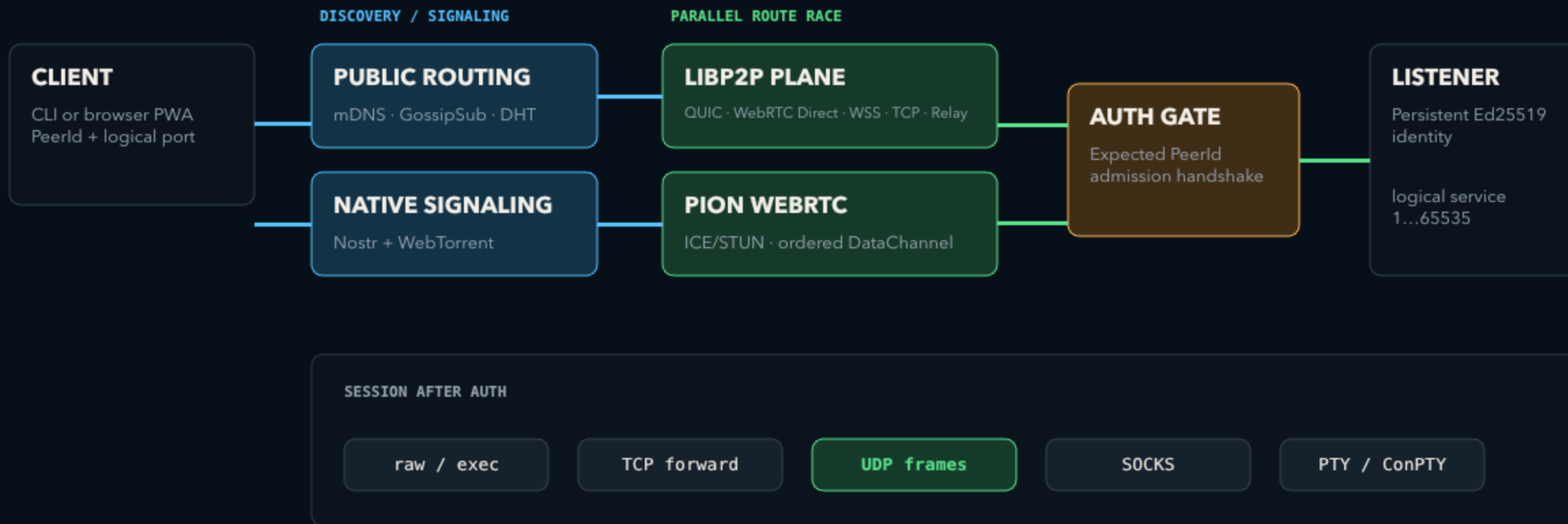
## P2P-NETCAT

### Identity stays stable



PeerId selects the right peer; discovery and route racing select a reachable path.

# The first route to complete authentication wins.



# Relay-only keeps the PeerId local – and makes ‘no direct path’ enforceable.

## DEFAULT ROUTING

**Fastest authenticated route wins**



Direct and relayed candidates can race. Reachability is maximized, but the host may expose a direct route.

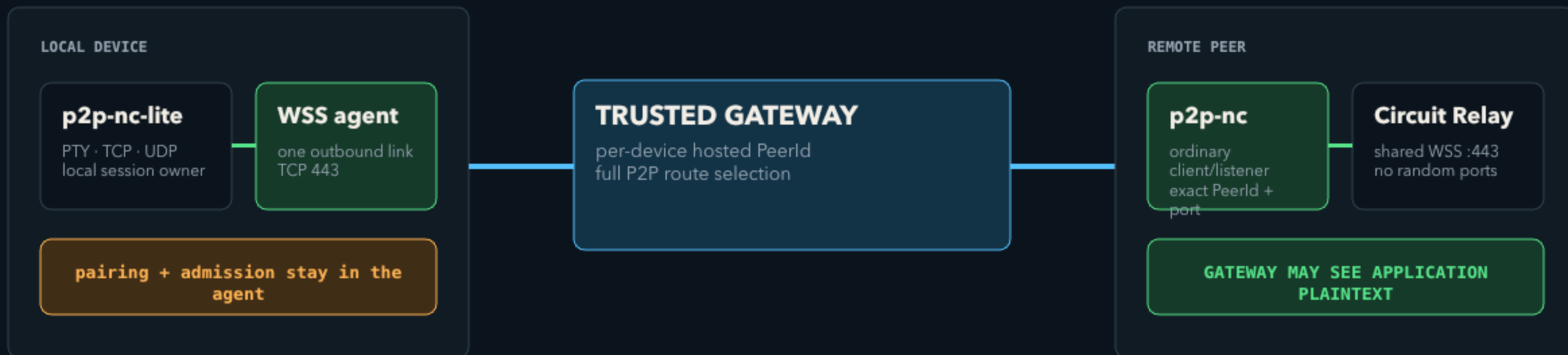
## STRICT RELAY-ONLY

**Only explicit Circuit Relay routes exist**



Requires an explicit relay; disables DHT, mDNS, PubSub, direct QUIC/WebRTC and fails closed if the relay is unavailable.

# A trusted WSS gateway moves the P2P host – and the trust boundary – to the VPS.



ONE FIXED PORT

agent opens WSS



gateway hosts PeerId



P2P stream reaches peer

Data, EOF, reset, and flow credit are multiplexed; local -i still starts the PTY on the device.

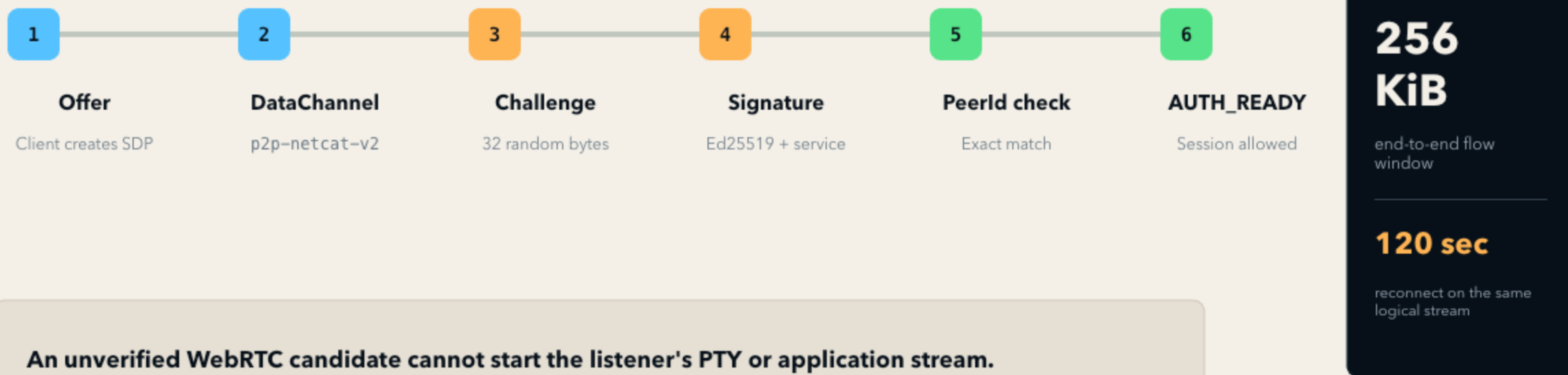


# One CLI covers terminals, streams, datagrams, and proxies.

| MODE       | LOCAL INTERFACE    | REMOTE ACTION        | WHAT IS PRESERVED | FLAG  |
|------------|--------------------|----------------------|-------------------|-------|
| RAW / EXEC | stdin/stdout       | stream or command    | byte stream       | -e    |
| TCP FWD    | 127.0.0.1:PORT     | dial TCP target      | connection        | -p    |
| UDP FWD    | 127.0.0.1:PORT/udp | connected UDP socket | packet boundaries | -u -p |
| SOCKS      | SOCKS4/4a/5        | CONNECT remotely     | remote DNS        | -S    |
| PTY        | interactive TTY    | Unix PTY / ConPTY    | resize + EOF      | -i    |
| LISTENER   | logical port       | multiple sessions    | identity + policy | -l -k |

Wire protocol compatibility     /p2p-netcat/1.0.0/<port>   ·   /p2p-netcat/udp/1.0.0/<port>

# Application bytes flow only after the expected PeerId is proven.

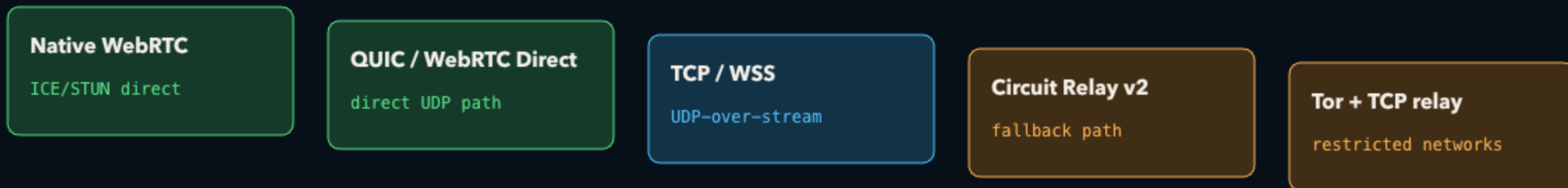


# UDP packet boundaries survive even over TCP/WSS/relay.

## DATAGRAM ASSOCIATION

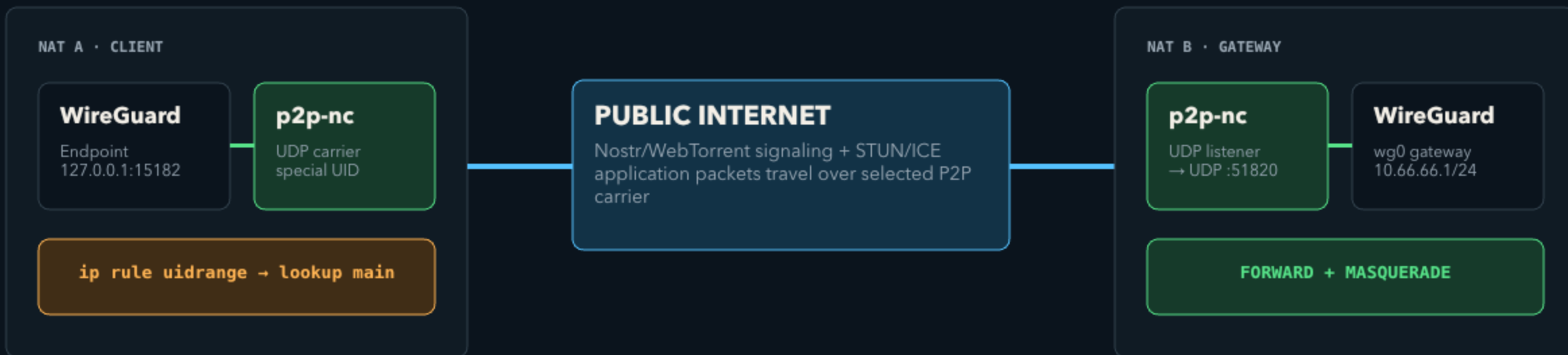


## FALLBACK LADDER



Fallback cost: ordered delivery can introduce head-of-line blocking.

# The carrier stays outside the default route, so the tunnel cannot consume itself.



## E2E CHECK

two NAT namespaces



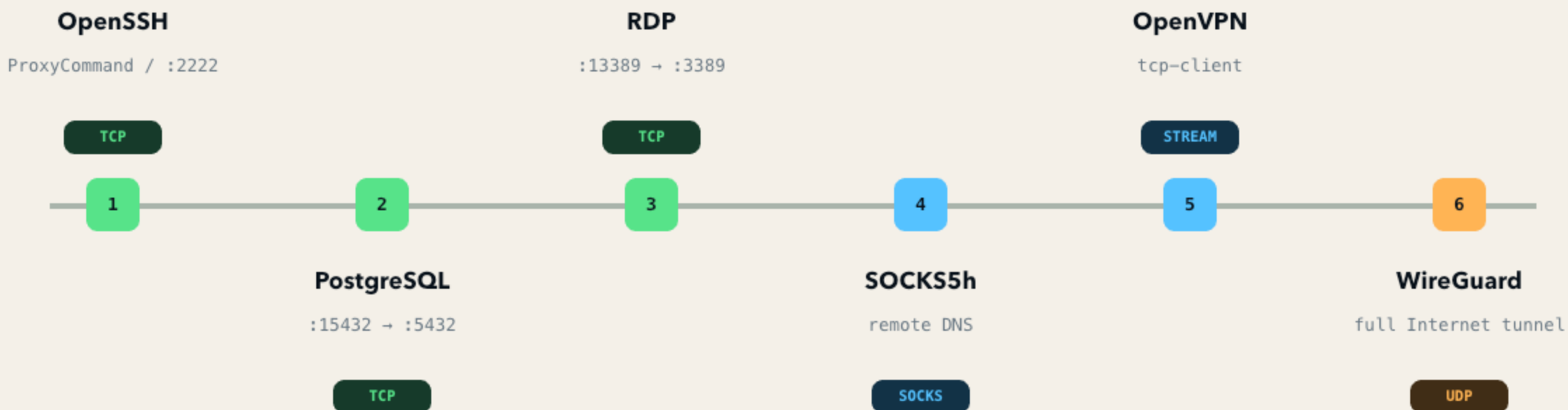
WireGuard handshake



HTTP egress = full-tunnel-ok

Start the carrier first, then wg-quick up wg0 – policy routing protects DNS, signaling, STUN, and reconnect.

# One primitive operation powers many working tools.



The logical port selects the P2P service; it does not have to match the local TCP/UDP port.

# One installation command; four delivery surfaces.

## VERIFIED DEPLOY

```
curl -fsSL .../deploy.sh | bash
```

- detects OS + architecture
- downloads the release
- verifies SHA256SUMS
- installs p2p-nc + pnc

## RELEASE BINARIES

Linux · macOS · Windows · Android  
amd64 / arm64

## GHCR CONTAINER

ghcr.io/santaklouse/go-p2p-netcat  
non-root · /config identity

## GO INSTALL

cmd/p2p-nc + cmd/pnc  
CGO is not required

## BROWSER PWA

## Static

GitHub Pages

React UI  
Web Worker  
IndexedDB routes  
Native WebRTC

# Tests target the exact failure boundaries introduced by the architecture.

## NATIVE WEBRTC SOAK

**60 / 60**

workload iterations

**1 152 MiB**

transferredBytes in a fresh JSON report

Pion DataChannel · SHA-256 · reconnect

## CI COVERAGE

### Go tests

Linux · macOS · Windows

### Race detector

Linux · ./...

### Full tunnel

2 NAT + WireGuard + HTTP

### Docker

buildx + runtime checks

### WebRTC soak

weekly Ubuntu + macOS

## WHAT SOAK DOES NOT PROVE

- public Nostr/tracker availability
- intercontinental latency
- every NAT and firewall type
- background browser tabs and sleep
- 100% relay-free connectivity

# Three command pairs cover the highest-value use cases.

Every example uses the documentation PeerId and real ports.

## SSH THROUGH A LOCAL PORT

```
server$ p2p-nc -l -k -d 127.0.0.1 -p 22 22022
client$ p2p-nc -p 2222 12D3KooWQ3uxpHgjdKE6vGmvzKS8RPbxUDLwJ7XCLaD6YXdUfbR9 22022
client$ ssh -p 2222 alice@127.0.0.1
```

## TCP PORT FORWARDING

```
server$ p2p-nc -l -k -d 127.0.0.1 -p 5432 25432
client$ p2p-nc -p 15432 12D3KooWQ3uxpHgjdKE6vGmvzKS8RPbxUDLwJ7XCLaD6YXdUfbR9 25432
```

## WIREGUARD CARRIER FOR A FULL TUNNEL

```
server$ p2p-nc -u -l -k -d 127.0.0.1 -p 51820 35182
client$ sudo wireguard-full-tunnel.sh -- /usr/local/bin/p2p-nc -u --udp-idle-timeout 0 \
      -p 15182 12D3KooWQ3uxpHgjdKE6vGmvzKS8RPbxUDLwJ7XCLaD6YXdUfbR9 35182
client$ sudo wg-quick up wg0
```



# Use p2p-netcat when identity should outlive the address.

PeerId is the stable entry point. Route racing and fallback are the delivery mechanism.

[github.com/santaklouse/go-p2p-netcat](https://github.com/santaklouse/go-p2p-netcat)

## GOOD FIT

- access to home and edge hosts
- SSH and private TCP services
- UDP/WireGuard across NAT
- temporary tunnels without a control plane

## PLAN B REQUIRED

- symmetric NAT or blocked UDP
- guaranteed enterprise ingress
- unsupported SOCKS UDP ASSOCIATE

TURN · Circuit Relay v2 · reachable TCP/WSS route